

Beyond Success and Violation: Manipulation-Entry Collapse in Safe Robotic Reinforcement Learning under Distribution Shift

Anonymous Author(s)

Abstract

Aggregate success and safety-violation rates are standard in safe robotic reinforcement learning, yet under distribution shift in contact-rich manipulation they can hide a distinct failure mode: policies remain nominally safe but lose access to task-relevant interaction trajectories before sustained manipulation.

It is well understood that safety constraints can induce conservative behavior; however, it remains unclear whether constrained fine-tuning under shift primarily degrades post-contact competence or reduces manipulation accessibility—the probability of entering contact-rich manipulation regimes at all. We study this question in robosuite Lift (Panda, controlled object-pose perturbations) using warm-started Lagrangian constrained fine-tuning from an unconstrained SAC-Value prior, including phase-aware M4.1, an instantaneous-force baseline, and a global- λ variant.

We find manipulation-entry collapse: on a matched-seed robosuite Lift grid, constrained policies retain comparatively stable conditional success after entry, but lose manipulation-entry probability under pose shift while violation rates can remain low across constraint variants. Violation rates can stay low even as entry falls, so aggregate metrics may overestimate robustness. We advocate phase-resolved evaluation that separates manipulation-entry from post-entry success alongside standard safe-RL reporting. Extended experiments, matrices, and reproducibility materials are in the accompanying supplement and multimedia repository.

1. Introduction

Reinforcement learning has strong potential for contact-rich robotic manipulation, yet safe execution remains difficult because task progress and physical interaction risk are tightly coupled. A policy must approach an object, establish contact, regulate forces, and com-

plete lifting while respecting safety limits—often under sparse rewards and discontinuous contact dynamics.

Safe manipulation is typically evaluated with aggregate task success and constraint-violation rates. These metrics are convenient, but they compress an entire rollout into a few scalars and cannot distinguish where a policy fails along the interaction timeline.

It is well understood that safety constraints can induce conservative behavior [1, 2]: policies may slow down, avoid uncertain contacts, or terminate early to keep costs low. What remains less clear under *distribution shift* is whether constrained fine-tuning primarily degrades *post-contact* control quality, or instead reduces the probability of entering manipulation-relevant interaction in the first place. The two outcomes imply different failure mechanisms and different remediation strategies, yet they are often conflated when only terminal success and violation rates are reported.

We therefore ask: for warm-started constrained fine-tuning, does geometric shift mainly reduce manipulation *accessibility* or post-entry competence? We study this question in robosuite Lift (Panda manipulator) as a controlled case study. Our goal is not a new safe-RL algorithm, but a mechanistic account of how phase-aware Lagrangian shaping changes reachable behavior when object pose is perturbed at test time.

We use a standard two-stage pipeline: an unconstrained SAC-Value capability prior (2M steps), followed by 200k-step constrained fine-tuning with bounded per-phase dual variables and EMA-smoothed force costs (M4.1). Under nominal conditions, constrained policies achieve competitive safe manipulation. Under shift, performance loss is not primarily explained by higher violation rates; instead, failures concentrate before sustained manipulation entry, while conditional success given entry remains comparatively stable. We term this pattern *manipulation-entry collapse* (Fig. 1).

Matched-seed ablations with instantaneous-force (M4) and global- λ baselines exhibit the same entry-reduction pattern, indicating that the phenomenon in our testbed is associated with constrained shaping rather than a single implementation knob (Sec. 5.3).

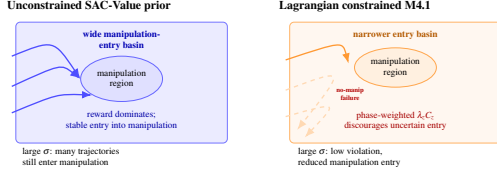


Figure 1. Schematic incentive structure (not measured trajectories). Solid: rollouts that reach manipulation; dashed: failure before entry. The SAC-Value prior maintains a broad manipulation-entry basin. Phase-weighted safety costs can narrow the basin so that, under shift, avoiding entry becomes lower-cost than recovering after contact.

Paper organization. Sec. II reviews safe RL, manipulation robustness, and evaluation gaps. Sec. III presents the phase CMDP fine-tuning pipeline and entry-resolved metrics. Sec. IV describes the Lift benchmark and shift protocol. Sec. V reports nominal/robustness results, mechanism analyses, and attributions; Sec. VI discusses implications and limitations.

Contributions:

- **Phenomenon.** We identify manipulation-entry collapse under shift: reduced manipulation-entry probability with comparatively stable $P(\text{succ}|\text{enter})$.
- **Mechanism.** Phase-resolved failure decomposition and transition analyses show losses concentrate before manipulation entry, not only in post-contact control.
- **Implications.** Low violation rates can coexist with reduced entry; we advocate reporting manipulation-entry and post-entry metrics alongside aggregate safe-RL scores.

2. Related Work

Safe and constrained reinforcement learning. Safe RL is commonly formulated as a constrained Markov decision process (CMDP), maximizing return subject to expected safety costs [3]. Surveys distinguish objective shaping, exploration restriction, and Lagrangian formulations [1, 4]. Trust-region and primal-dual methods such as CPO [2] and Lyapunov-based approaches [5] scale to continuous control; extensions address unknown safe sets and state-wise constraints [6, 7].

Our work is complementary to these algorithmic lines: we do not propose a new safe-RL optimizer, but study how Lagrangian constrained *fine-tuning* reshapes

the reachable behavior of an already-capable manipulation policy. Most prior evaluations emphasize aggregate return and constraint return. They rarely ask whether a deployable policy still *enters* contact-rich manipulation phases after warm-started fine-tuning when test conditions shift—the gap we target with entry-resolved metrics.

Robustness and manipulation. Manipulation is inherently contact-rich: success depends on reaching and sustaining interaction phases, not only on terminal lift labels [8–10]. Recent work targets recovery from out-of-distribution states back to the training manifold [11], separating accessibility of task-relevant states from headline success. Deep RL policies trained in simulation can still be sensitive to small pose or timing perturbations [12, 13]. Our study uses controlled Gaussian xy object noise in Lift to isolate how *constrained* fine-tuning changes phase accessibility rather than proposing another robustness algorithm.

Evaluation gaps. Robotics failure analyses often categorize planning, control, and contact faults, but safe-RL papers seldom decompose failures into *before* versus *after* manipulation entry. Aggregate violation rate alone can decrease when a policy avoids contact altogether—a policy that rarely grasps can look safe while failing the manipulation task. Recent manifold-recovery work [11] separates accessibility of task-relevant states from headline success; we apply the same spirit to constrained RL by elevating manipulation-entry rate, $P(\text{succ}|\text{enter})$, and phase-transition statistics to the same reporting level as success and violation in contact-rich settings.

3. Method

3.1. Phase-aware constrained fine-tuning

We model manipulation as a phase-augmented CMDP. At each timestep the environment assigns a label $z_t \in \{\text{free}, \text{pre}, \text{contact}, \text{manip}\}$ from gripper-object distance, contact geometry, gripper closure, and grasp/success signals (labels are *not* policy inputs). Safety is enforced through phase-specific cost budgets,

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_t r_t \right] \text{ s.t. } \mathbb{E}_{\pi} \left[\sum_t c_{z_t}(s_t, a_t) \right] \leq \varepsilon_{z_t}. \quad (1)$$

Intuitively, **free** denotes far approach; **pre** near-object approach; **contact** close range, finger contact, or closing; **manip** grasped object or task success. A two-step hysteresis filter reduces label chatter at phase boundaries (threshold details in Supplement Sec. A).

A SAC-Value policy—stochastic π , twin Q , and state value V —is pretrained for 2M steps without safety constraints, minimizing

$$\mathcal{L}_\pi^{\text{prior}} = \mathbb{E}[\alpha \log \pi(a|s) - (\min_i Q_{\phi_i}(s, a) - V_\psi(s))]. \quad (2)$$

We then fine-tune for 200k steps with per-phase cost critics C_z and bounded dual variables $\lambda_z \in [0, \lambda_{\max}]$ ($\lambda_{\max}=20$). The actor minimizes

$$\mathcal{L}_\pi = \mathbb{E}[\alpha \log \pi(a|s) - (\min_i Q_{\phi_i}(s, a) - V_\psi(s)) + \lambda_z C_z(s)]. \quad (3)$$

Per-phase costs include velocity (free), gripper closing rate (pre), and EMA-smoothed sustained contact force (contact/manip). Phase budgets ($\epsilon_{\text{free}}, \epsilon_{\text{pre}}, \epsilon_{\text{contact}}, \epsilon_{\text{manip}}$) = (0.02, 0.03, 0.08, 0.12) increase toward manipulation to reflect tighter tolerances during lifting.

EMA sustained force cost. Let F_t denote gripper-object contact force and $\bar{F}_t = \alpha_F \bar{F}_{t-1} + (1 - \alpha_F) F_t$ with $\alpha_F=0.85$. Contact/manip costs activate on sustained exceedance of force thresholds rather than single-timestep spikes, reducing penalty on brief grasp transients while penalizing persistent unsafe contact (thresholds in Supplement Sec. A).

Dual updates. Each λ_z is updated from batch estimates of phase cost return and projected to $[0, \lambda_{\max}]$. Intuitively, phases with sustained cost violations increase λ_z , strengthening the actor penalty for that phase until costs fall below ϵ_z . This is standard primal-dual CMDP fine-tuning; our contribution is not the optimizer but the phase-resolved *evaluation* that reveals accessibility loss under shift.

3.2. Evaluation decomposition

We report four complementary metrics: (i) task success; (ii) safety violation rate and peak force $F_{\max}$; (iii) manipulation-entry rate, with

$$\text{manip_entered} = \mathbf{1}\{\text{manip_frac} > 0.05\}; \quad (4)$$

(iv) conditional success $P(\text{succ}|\text{enter})$. Failed episodes are labeled *no-manipulation* (never enter) or *post-manipulation* (fail after entry). Phase-transition rates $P(\text{to}|\text{from})$ over $\{\text{free}, \text{pre}, \text{contact}, \text{manip}\}$ diagnose whether shift blocks approach and contact transitions or post-contact execution (Supplement Sec. D). This decomposition is the main analytic tool for separating accessibility from post-contact competence.

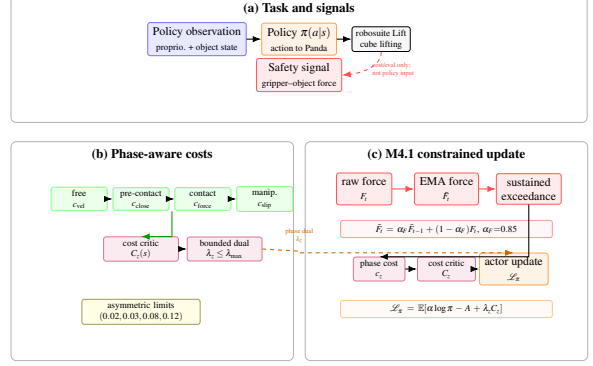


Figure 2. Overview: unconstrained SAC-Value prior, phase-conditioned CMDP fine-tuning (M4.1), and evaluation under controlled pose shift.

Compared policies. We evaluate four constrained-training variants against the same SAC-Value prior: (i) M4.1 baseline and replication (repl0); (ii) M4 warm-start with instantaneous-force costs and without EMA sustained penalties; (iii) global- λ fine-tuning that replaces per-phase λ_z with one shared multiplier while keeping the same phase costs and EMA force treatment. All arms share the Lift simulator, observation pipeline, and evaluation protocol, isolating constraint shaping rather than environment differences.

4. Experimental Setup

We evaluate on robosuite Lift with a Panda arm (500 control steps per episode, 20 Hz). The agent must approach, contact, grasp, and lift a cube above a height threshold. Observations include proprioception and object state; gripper-object force is logged for safety costs and evaluation but is *excluded* from the policy input, matching deployable sensing assumptions in many manipulation stacks.

Distribution shift protocol. After each reset, the cube initial (x, y) position is perturbed as $(x, y) \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma \in \{0, 1, 2, 4, 6\}$ cm. The perturbation is a controlled geometric shift in object pose, not a relabeling of random seeds. All policies are evaluated with deterministic action selection. Each policy- σ pair uses 30 episodes; the main robustness grid shares reset indices across methods at every σ so differences reflect policy behavior rather than reset luck. Wilson 95% confidence intervals are computed from per-episode logs (Supplement Sec. H).

Table 1. Nominal evaluation ($\sigma=0$; 30 episodes; deterministic).

Method	Success	Manip. frac	Viol.	Mean $F_{\max}$
M4 warm-start	27%	—	—	~ 59 N
M4.1 baseline	60%	0.66	2.5%	41.5 N
M4.1 repl0	67%	0.73	2.5%	53.5 N

Table 2. Main robustness grid at $\sigma=6$ cm (30 ep.; matched seeds).

Method	Success	Entry	$P(s e)$	Viol.
SAC-Value prior	63.3%	86.7%	73.0%	4.9%
M4.1 baseline	33.3%	53.3%	62.5%	1.7%
M4.1 repl0	40.0%	53.3%	75.0%	2.5%

Training. Training follows a two-stage pipeline (Method): 2M-step unconstrained SAC-Value pretraining, then 200k-step constrained fine-tuning with warm-started actor and task-reward critics. Phase cost critics and Lagrange multipliers are trained during fine-tuning. We use a standard SAC backbone (MLP [256,256], $\gamma=0.99$, learning rate 3×10^{-4} , batch 256, replay size 10^6 ; full hyperparameters in Supplement Sec. A). Safety violation at evaluation is defined as any timestep with $F_{\max}$ above the environment force limit; peak force is the episode maximum.

Metrics logged per episode. Beyond success and violation, each rollout logs manipulation fraction (time in manip), manip_entered (Eq. (4)), phase occupancy, $F_{\max}$, and failure labels (no-manipulation vs. post-manipulation). These fields are computed offline from the same simulator traces used for training logs.

Compared policies. The main grid compares the SAC-Value prior, M4.1 baseline, and an independent M4.1 replication (repl0). Attribution cohorts additionally include M4 warm-start (instantaneous-force costs) and global- λ fine-tuning (single shared multiplier).

Table 1 summarizes nominal ($\sigma=0$) performance after warm-start. Table 2 previews the $\sigma=6$ cm grid analyzed in Sec. 5.1.

5. Results

5.1. Nominal and robustness performance

Nominal safe manipulation. After warm-starting from the unconstrained prior, M4.1 reaches 60–67% success with manipulation fraction 0.66–0.73 and $\sim 2.5\%$ violation (Table 1). M4 warm-start under the same evaluation protocol reaches only 27% success, indicating that phase-aware EMA force costs and per-

phase budgets materially affect nominal capability in this testbed. These results establish that constrained fine-tuning can be task-feasible under unperturbed re-sets before testing shift.

Robustness curves. Figure 3 reports success, manipulation-entry rate, and $P(\text{succ}|\text{enter})$ across σ . All methods degrade as σ increases, but the curves decouple: the prior loses success more gradually and retains higher entry at large σ , while constrained policies show a steeper drop in entry with comparatively flat conditional success after entry. At $\sigma=6$ cm (Table 2), the prior achieves 63.3% success and 86.7% entry versus 33–40% success and 53.3% entry for M4.1, while $P(\text{succ}|\text{enter})$ remains 62.5–75.0% for constrained arms versus 73.0% for the prior. Intermediate noise levels ($\sigma=2$ –4 cm) show the same decoupling: entry drops earlier than conditional success for constrained policies (Fig. 3; full grid in Supplement Sec. C).

Takeaway: constrained fine-tuning preserves nominal safety but loses robustness primarily through reduced manipulation entry, not through uniformly worse post-contact success.

5.2. Mechanism analysis of entry collapse

We next ask *where* failures occur along the phase timeline.

Failure decomposition. Figure 4 summarizes failed episodes at $\sigma=6$ cm on the main grid. Among M4.1 failures, 14/20 baseline and 14/18 repl0 episodes end before sustained manipulation, whereas 7/11 SAC-Value failures occur after manipulation entry. The asymmetry indicates that constrained policies disproportionately fail by never reaching manipulation, while the prior more often fails during post-entry execution.

Phase transitions. Figure 5 reports empirical $P(\text{to}|\text{from})$ at $\sigma=0$ and 6 cm. At $\sigma=6$ cm, M4.1 shows lower pre $\rightarrow$ contact flow and lower manipulation-phase occupancy ($\approx 38\%$ vs. 67% for the prior). Once contact is established, contact $\rightarrow$ manip rates remain closer across methods. Full 4×4 matrices appear in Supplement Sec. D.

Qualitative behavior. Matched-seed rollouts (multi-media video) show M4.1 completing lifts at $\sigma=0$ cm but often hesitating near the perturbed cube at $\sigma=6$ cm, with larger standoff and fewer contact attempts, while the prior still commits to grasping. This visual pattern aligns with reduced entry rather than unstable post-contact control in the trajectories we inspected.

Robustness under controlled object-pose perturbation; shaded bands: Wilson 95% CI

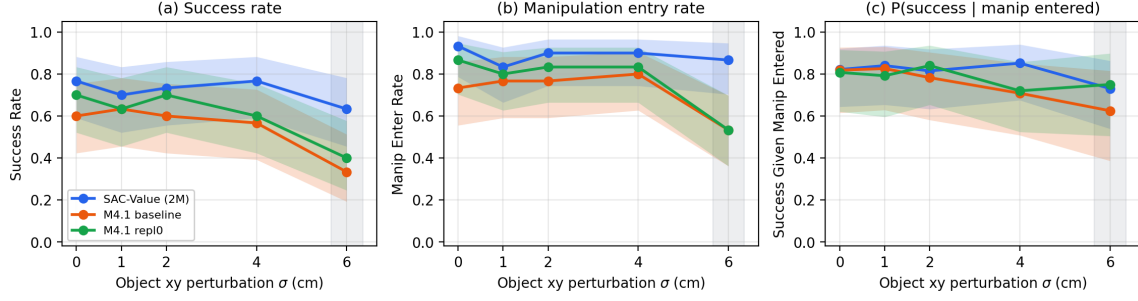


Figure 3. Robustness under cube xy perturbations (σ in cm; 30 episodes per cell; Wilson 95% CI). Main grid: SAC-Value prior vs. M4.1 baseline and repl0.

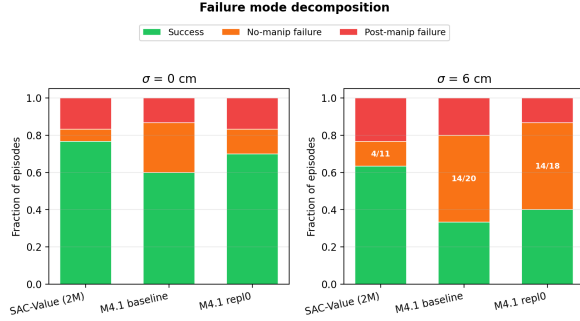


Figure 4. Failure modes among failed episodes at $\sigma=6$ cm (main grid).

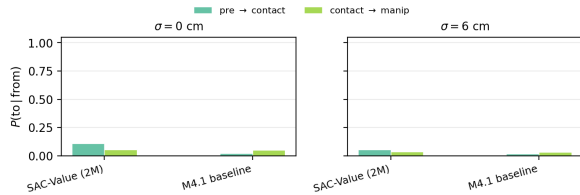


Figure 5. Phase-transition rates at $\sigma=0$ (left) and 6 cm (right; matched seeds).

Takeaway: failures concentrate before manipulation entry; post-contact transitions are comparatively preserved.

5.3. Attribution and safety tradeoff

Across constraint formulations. Table 3 compares the SAC-Value prior, M4 warm-start, and M4.1 on a matched-seed attribution cohort at $\sigma=6$ cm. All constrained arms reduce entry relative to the prior (40–56.7% vs. 83.3%) while $P(\text{succ}|\text{enter})$ stays in a similar

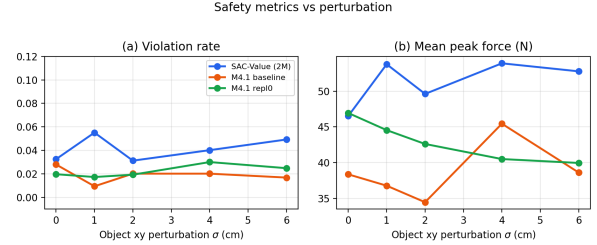


Figure 6. Violation rate and mean peak force vs. σ (main grid).

band (58–84%). Table 4 shows the same entry reduction for global- λ versus phase-aware λ_z at $\sigma=6$ cm (53.3% entry for both constrained arms vs. 86.7% for the prior), with different success but similar accessibility.

Violations vs. entry. On the main grid at $\sigma=6$ cm, M4.1 violation rates remain low (1.7–2.5%) with mean peak force ≈ 39 –40 N, while the prior reaches higher success but higher violation (4.9%) and force (52.8 N). Low violation therefore *coexists* with low success for constrained policies in these runs—consistent with contact avoidance rather than uniformly unsafe post-contact behavior. Figure 6 shows that violation and force stay relatively flat for M4.1 as σ grows while success and entry fall, reinforcing that aggregate safety metrics can mask entry collapse.

Takeaway: entry collapse is not tied to a single constraint implementation and can be masked by aggregate safety metrics.

Table 3. Attribution cohort at $\sigma=6$ cm (30 ep., matched seeds).

Method	Succ.	Entry	$P(s e)$	Viol.
SAC-Value prior	70.0%	83.3%	84.0%	5.0%
M4 warm-start	23.3%	40.0%	58.3%	2.8%
M4.1	46.7%	56.7%	82.4%	2.4%

Table 4. Phase λ_z vs. global λ at $\sigma \in \{0, 6\}$ cm.

Method	σ	Succ.	Entry	$P(s e)$
SAC-Value prior	0	76.7%	93.3%	82.1%
SAC-Value prior	6	63.3%	86.7%	73.1%
M4.1	0	60.0%	73.3%	81.8%
M4.1	6	33.3%	53.3%	62.5%
Global λ	0	66.7%	83.3%	80.0%
Global λ	6	36.7%	53.3%	68.8%

6. Discussion

6.1. Interpretation under constrained shaping

Our results are consistent with a simple incentive story (Fig. 1). Under shift, misaligned object pose increases the expected cost of exploratory contact. Phase-weighted Lagrangian penalties make *avoiding* manipulation a lower-cost response than recovering after imperfect contact, even when the prior would still commit to grasping. This is not a claim that all safe-RL methods must collapse entry; rather, in our warm-started fine-tuning setting, aggregate violation can improve while accessibility to manipulation phases degrades.

6.2. Reporting recommendations

We recommend that contact-rich manipulation papers report, at minimum: (i) task success; (ii) violation rate and a force statistic such as $F_{\max}$; (iii) manipulation-entry rate (Eq. (4)); (iv) $P(\text{succ}|\text{enter})$; and (v) a failure-mode split between no-manipulation and post-manipulation episodes. When feasible, phase-transition rates $P(\text{to}|\text{from})$ help localize whether shift blocks approach, contact, or post-contact execution. These quantities are inexpensive to log from the same rollouts used for safe-RL benchmarks and make entry collapse visible before deployment.

6.3. Alternative explanations considered

Supplement Sec. E maps hypotheses to falsifiable predictions. Briefly: (a) if failure were primarily post-contact control degradation, we would expect $P(\text{succ}|\text{enter})$ to drop sharply under shift while entry remains high—not observed on the main grid; (b) if violations dominated, constrained policies would

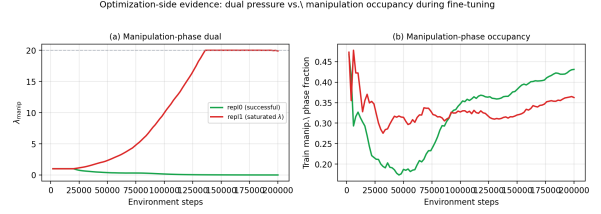


Figure 7. Training-time λ_{manip} vs. manipulation fraction (fine-tuning).

show rising violation under shift while still attempting manipulation—violation stays low while entry falls; (c) if the effect were seed-specific optimization noise only, replications would disagree on entry trends—both M4.1 replications show reduced entry at $\sigma=6$ cm. Training-time λ_{manip} trajectories (Fig. 7; Supplement Sec. B) show that dual variables can rise during fine-tuning even as manipulation fraction fluctuates, motivating joint reporting of optimization and behavioral metrics.

6.4. Scope and limitations

Evidence is simulation-only on robosuite Lift with Gaussian xy pose noise; contact physics, sensing noise, and human-scale perturbations may interact differently with constrained shaping. We study warm-started fine-tuning from a strong prior rather than training safe policies from scratch. Hardware validation, alternative constraint parameterizations, and objectives that explicitly preserve manipulation accessibility (e.g., entry-aware budgets or contact-encouraging shaping during shift) remain important next steps.

7. Conclusion

We studied manipulation-entry collapse in warm-started safe fine-tuning for contact-rich lifting. Across nominal and shifted evaluations in robosuite Lift, constrained policies can achieve strong safe performance at $\sigma=0$ yet, under pose perturbation, frequently fail before entering manipulation. Conditional success given entry remains comparatively stable, and the pattern appears across M4.1, M4, and global- λ variants.

These results highlight an evaluation blind spot: aggregate success and violation rates can agree that a policy is “safe” while it rarely engages in task-relevant manipulation under shift. We recommend reporting manipulation-entry rate, $P(\text{succ}|\text{enter})$, and phase-transition or failure-mode breakdowns alongside conventional safe-RL metrics.

Limitations. Evidence is simulation-only on a sin-

gle task with Gaussian xy shift; contact timing, orientation noise, and sim-to-real gaps may change which phases fail first. We study warm-started fine-tuning rather than safe learning from scratch, and do not claim optimality of the chosen phase budgets or EMA parameters. Extended robustness grids, Fisher-style comparisons, full transition matrices, statistical tests, and reproducibility artifacts are in the accompanying supplement; qualitative rollouts are in the multimedia attachment.

learning,” in *arXiv preprint arXiv:2009.12293*, 2020.

References

- [1] J. García and F. Fernández, “A comprehensive survey on safe reinforcement learning,” *Journal of Machine Learning Research*, vol. 16, pp. 1437–1480, 2015.
- [2] J. Achiam, D. Held, A. Tamar, and P. Abbeel, “Constrained policy optimization,” *International Conference on Machine Learning*, 2017.
- [3] E. Altman, *Constrained Markov Decision Processes*. Chapman and Hall/CRC, 1999.
- [4] A. Wachi, X. Shen, J. Simão, and Y. Sui, “A survey of constraint formulations in safe reinforcement learning,” in *International Joint Conference on Artificial Intelligence*, 2024.
- [5] Y. Chow, O. Nachum, M. Ghavamzadeh, P. Daunou, and P. Abbeel, “Lyapunov-based safe policy optimization,” *arXiv preprint arXiv:1805.07785*, 2018.
- [6] A. Wachi and Y. Sui, “Safe reinforcement learning in constrained markov decision processes,” in *International Conference on Machine Learning*, 2020.
- [7] W. Zhao, T. He, C. Liu, and P. Abbeel, “State-wise safe reinforcement learning: A survey,” in *International Joint Conference on Artificial Intelligence*, 2023.
- [8] Z. Su, O. Kroemer, A. Englebert, Y. Duan, S. Schaal, and G. Sukhatme, “Learning manipulation graphs from demonstrations using multimodal sensory signals,” in *IEEE International Conference on Robotics and Automation*, 2018.
- [9] K. Takeuchi, S. Sakaino, and T. Tsuji, “Motion generation based on contact state estimation using two-stage clustering,” *IEEJ Journal of Industry Applications*, 2023.
- [10] F. Zhu, L. Wang *et al.*, “Failure handling of robotic pick and place tasks with multimodal cues under partial object occlusion,” *Frontiers in Neurorobotics*, 2021.
- [11] A. Reichlin, G. Marchetti *et al.*, “Back to the manifold: Recovering from out-of-distribution states,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022.
- [12] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in *International Conference on Machine Learning*, 2018.
- [13] Y. Zhu, J. Wong, A. Mandlekar, R. Martín-Martín, A. Joshi, S. Nasiriany, and Y. Zhu, “robosuite: A modular simulation framework and benchmark for robot